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Defence against the modern arts: the curse of statistics: Part I—FRStat

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For several decades, legal and scientific scholars have argued that conclusions from forensic examinations should be supported by statistical data and reported within a probabilistic framework. Multiple models have been proposed to quantify and express the probative value of forensic evidence. Unfortunately, the use of statistics to perform inferences in forensic science adds a layer of complexity that most forensic scientists, court offices and lay individuals are not armed to handle. Many applications of statistics to forensic science rely on *ad hoc* strategies and are not scientifically sound. The opacity of the technical jargon that is used to describe these probabilistic models and their results, and the complexity of the techniques involved make it very difficult for the untrained user to separate the wheat from the chaff. This series of article is intended to help forensic scientists and lawyers recognize limitations and issues in tools proposed to interpret the results of forensic examinations. This article focuses on the tool proposed by the Latent Print Branch of the U.S. Defense Forensic Science Center (DFSC) and called FRStat. In this article, I explore the compatibility of the results outputted by FRStat with the language used by the DFCS to report the conclusions of their fingerprint examinations, as well as the appropriateness of the statistical modelling underpinning the tool and the validation of its performance.

Keywords: FRStat; fingerprint; statistics; validation.

1. Introduction

For more than half a century, legal and scientific scholars have argued that conclusions from forensic examinations should be supported by statistical data and reported within a probabilistic framework. Unfortunately, the use of statistics to perform inferences in forensic science adds a layer of complexity to forensic examinations that most forensic scientists, court officers and lay individuals are not armed to handle. There is no doubt that data, statistical analyses and probabilistic models are required to support all aspects of forensic examinations. However, there is a curse associated with the use of statistics and probability in forensic science (as well as in any other field): the opacity of the technical jargon used to describe probabilistic models and the results of statistical analyses, and the complexity of the techniques involved make it very difficult for the untrained user to separate the wheat from the chaff. As a result, a poor use of statistics and probability theory may go unnoticed, give an unwarranted varnish of legitimacy to a given forensic examination and unduly increase the level of trust that forensic scientists, court officers and lay individuals may associate with this examination.

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This article focuses on the changes that the Latent Print Branch of the U.S. Defense Forensic Science Center (DFSC) has recently introduced to their way of reporting fingerprint evidence and on the statistical tool that they are proposing to support their conclusions. These changes were made to account for numerous criticisms directed at the field of fingerprint examination over the past decades. In particular, in 2015, the DFSC issued an information paper where it announced its decision to move away from the categorical decision scheme of identification, exclusion and inconclusive examination. Instead, the Latent Print Branch proposed to use a more probabilistic language (U.S. Department of the Army, Defense Forensic Science Center, 2015):

> *The latent print on Exhibit ## and the record finger/palm prints bearing the name XXXX have corresponding ridge details. The likelihood of observing this amount of correspondence when two impressions are made by different sources is considered extremely low.*

This language was modified in 2017 to highlight that two alternative propositions must be considered when evaluating the weight of forensic evidence (U.S. Department of the Army, Defense Forensic Science Center, 2017):

> *The latent print on Exhibit ## and the standards bearing the name XXXX have corresponding ridge detail. The probability of observing this amount of correspondence is approximately ## times greater when impressions are made by the same source rather than by different sources.*

The DFSC has developed a statistical tool, FRStat (Swofford *et al*., 2018), which is intended to support fingerprint examiners using this language. This tool has been introduced in U.S. military court and has been offered in a recent civilian case in Texas (Ex Parte Areli Escobar, 2019).

In this article, I take an agnostic view on the type of conclusions that should be reported following fingerprint examinations. I note that the developers of FRStat make it very clear in their article (Swofford *et al*., 2018, p. 118) that they had no intention to develop a tool capable of assigning likelihood ratios, and I am not arguing that they should have. Instead, I explore several major issues associated with their project:

(1) There are critical discrepancies between the statistical objects that the FRStat algorithm produces, the logical and mathematical properties that they are assumed to possess and the way the DFSC proposes to report them.
(2) The statistical modelling underpinning FRStat calculation is inappropriate.
(3) The results of the validation of FRStat's performances are incorrectly interpreted.

In the next section, , I explain the algorithm behind FRStat and propose an appropriate interpretation of its outputs, and in Section 3, I describe the issues associated with the DFSC language and FRStat.

2. FRStat

FRStat is an algorithm, encapsulated in a software package, developed by the DFSC to support the probabilistic language announced in 2015 and 2017. FRStat was the subject of a publication by Swofford *et al*. (2018).[1]

[1] All of the content from Section 3 is interpreted from Swofford *et al*. (2018).

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2.1 *FRStat overview*

FRStat is based on two premises:

(1) Pairs of impressions from a given finger are more similar to each other than two impressions from two different fingers.
(2) The risk of making an erroneous identification increases when the level of similarity between a trace and a control impression decreases, while the risk of making an erroneous exclusion increases when the level of similarity increases.

Overall, FRStat provides a measure of the rates of erroneous identifications and exclusions that fingerprint examiners would have if they were making a decision of identification/exclusion at a given level of similarity between a trace and a control finger impression. Formally, FRStat is designed to test the following two hypotheses[2]:

H_0: the trace and control impressions originate from the same donor;

H_1: the trace and control impressions originate from two different donors.

FRStat works in a similar way to frequentist hypothesis testing. It can help examiners to accept or reject H_0 based on the probability of type I and type II errors. As such, FRStat relies on a test statistic, on the sampling distributions of that test statistic under the considered hypotheses and on the estimation of the type I and type II error rates for a given value of the test statistic using 'tail probabilities' assigned based on the sampling distributions. The two main differences between FRStat and the classic statistical hypothesis testing framework are that:

(1) the nomenclature of error types is inverted in the forensic context. FRStat's type I error rate (α) corresponds to the 'false negative rate or false exclusion rate' since erroneously rejecting H_0 results in wrongly concluding that the trace and control impressions originate from two different donors. FRStat's type II error rate (β) corresponds to the 'false positive rate or false identification rate' since failing to reject H_0 results in erroneously concluding that the trace and control impressions originate from the same donor;
(2) contrary to most statistical tests, FRStat estimates a rate of type II error: estimating the type II error enables accepting H_0 with a certain level of risk, instead of merely failing to reject it when the probability of type II error is unknown.

2.2 *FRStat's test statistic*

For a given comparison between a trace, t, and a control impression, c, FRStat's test statistic is based on the level of similarity, $d = \delta(t, c)$, between these impressions.[3] The manner in which FRStat measures this similarity is irrelevant for the rest of the discussion and is not studied any

[2] Note that the two hypotheses are not specific to a named individual. This point is important as it influences the design of the experiment(s) aimed at addressing the hypotheses. This type of hypotheses is considered to address the 'common source' identification problem in forensic science (Ommen *et al*., 2017; Ommen and Saunders, 2018).

[3] Here, the notation $\delta(\cdot, \cdot)$ represents some function that compares two fingerprints are returns a similarity score; upper case letters are used to denote the random variable (e.g. T, C), while lower case letters denote observed values for these random variables (e.g. t, c).

further. The only important point is that this function returns the level of similarity between any trace and control impressions as a 'continuous univariate' random variable (i.e. a real number). The scale and range of this number are equally irrelevant for the discussion below, apart from noting that larger numbers correspond to greater levels of similarity between two impressions.

2.3 *Sampling distributions*

To estimate the rates of erroneous identifications and exclusions for a given level of similarity and a given number of corresponding features, FRStat relies on the sampling distributions of the similarity statistic between any pair of trace and control impressions, $\delta(T, C)$, under the two hypotheses stated earlier, conditioned on the number of features. FRStat allows for processing fingerprint comparisons that have between 5 and 15 features. For each number of features, the sampling distributions result from an initial large experiment that involves the pairwise comparisons of a multitude of pairs of impressions known to originate from the same donor as each other (this experiment is supposed to reflect H_0, and the pairwise scores constitute the 'mated' dataset) and the pairwise comparisons of a multitude of pairs of impressions known to originate from different donors are each other (this experiment is supposed to reflect H_1, and the pairwise scores constitute the 'non-mated' dataset).

For each number of features, the 'mated' dataset enables estimating the rate of false exclusions (α) and comprises 1996 similarity scores sampled from 499 pairs of impressions obtained from 50 individuals (10 pairs from each individual). The 'mated' dataset for 15 features only includes 499 scores. For each number of features, the 'non-mated' dataset enables estimating the rate of erroneous identifications (β) and comprises 2000 similarity scores sampled from approximately 100 pairs of impressions, where the impressions in each pair are from two different donors.[4]

To extrapolate the rates of false positives and negatives beyond the range of observed similarity scores, FRStat uses parametric mixtures of logistic distributions to represent the 'probability densities' of the similarity scores, $\delta(T, C)$, in the mated and non-mated datasets.[5] The upper-left panel of Fig. 1 (Fig. 1a) represents these two probability densities. It illustrates the probability density for the scores between pairs of impressions in the mated dataset in light grey ($f_{\delta(T,C)}(\cdot|H_0)$ in Fig. 1a) and the probability density for the scores between pairs of impressions in the non-mated dataset in dark grey ($f_{\delta(T,C)}(\cdot|H_1)$ in Fig. 1a).

It is important to note that the distributions of scores in both datasets are fixed in the sense that they are not case-specific. The same distributions based on the same scores are pre-computed during the initial large experiment and used for all casework comparisons. These distributions are not

[4] It is not clear if Swofford *et al.* (2018) attempted to obtain non-mated scores as high as possible during the initial large experiment. Swofford *et al.* (2018, p. 115) states that '... the empirical distributions are intentionally biased such that the non-mated data are biased to higher similarity statistics values...'. In addition, in the supplemental appendix II of Swofford *et al.* (2018), which is referenced in section 2.2 of Swofford *et al.* (2018), we read that 'Each query print was then searched using an Automated Fingerprint Identification System (AFIS) against an operational database containing approximately 100 million different fingerprint impressions....' However, still in section 2.2, Swofford *et al.* (2018) state that 'The distributions of similarity statistic values ... of non-mated samples ... were generated using a subset of impressions from the ... NIST Special Database (SD) 27 ... randomly paired to non-mates'. My interpretation is that Swofford *et al.* (2018) consider scores from random non-mated pairs of impressions when building the distributions of the sampling statistics under H_1 but use the AFIS system to generate highly similar non-mated pairs of impressions when testing their model. Naturally, proper design of FRStat would have required the authors to study the non-mated distributions of the similarity statistic using highly similar non-mated impressions to begin with.

[5] More will be said about these mixtures of logistic distributions below.

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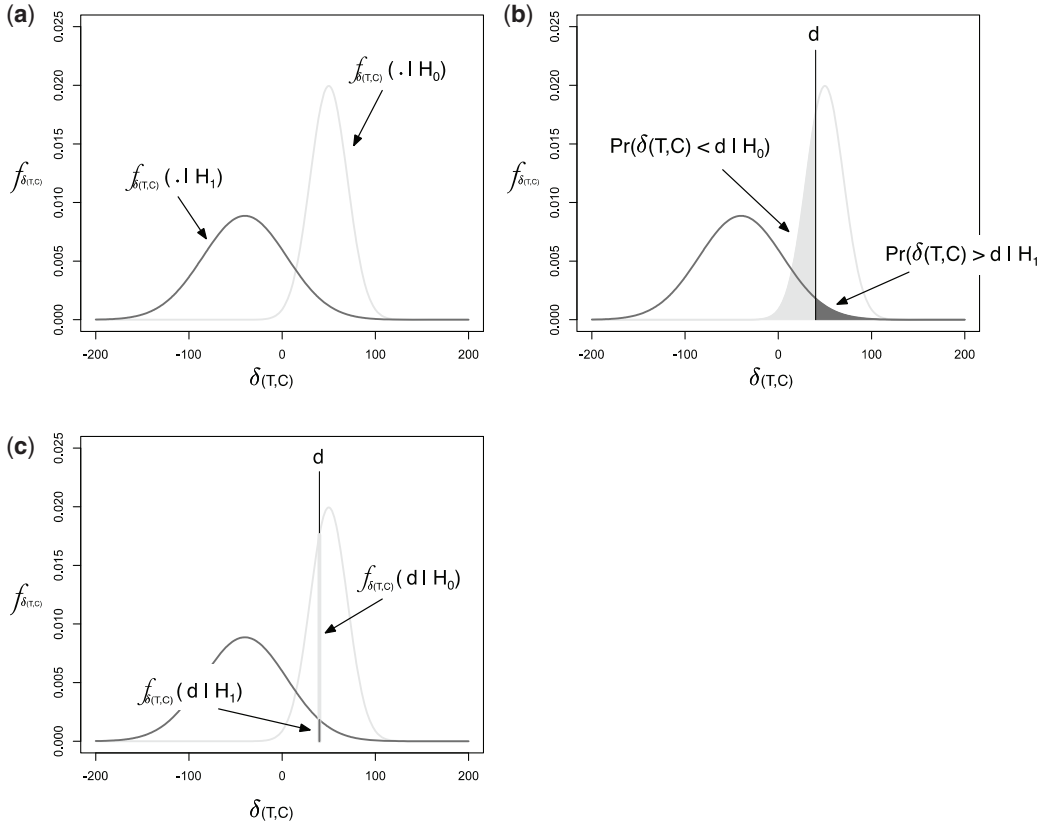


FIG. 1. Representation of the construction of FRStat ratios and score-based likelihood ratios. (**a**) This presents the two sampling distributions for the scores between pairs of impressions. Higher similarity levels result in higher scores. The distribution of scores from the same donor $f_{\delta(T,C)}(\cdot|H_0)$ is displayed in light grey, and the distribution of scores between pairs of impressions from different donors $f_{\delta(T,C)}(\cdot|H_1)$ is shown in dark grey. (**b**) This presents the 'tail probabilities' used by FRStat for an observed score, d, between a trace and control impression. The rate of false exclusions is defined as $\alpha = \Pr(\delta(T,C) < d|H_0)$ and is represented in light grey, while the rate of false identifications is defined as $\beta = \Pr(\delta(T,C) > d|H_0)$ and is represented in dark grey. (**c**) This represents a score-based likelihood ratio for the same observed score, d, as in (b). The score-based likelihood ratio is obtained by the ratio of the 'likelihoods' $f_{\delta(T,C)}(d|H_0)$ and $f_{\delta(T,C)}(d|H_1)$ represented in light and dark grey, respectively.

tailored to specific observations made in a given case. This implies that the rates of errors reported by FRStat are average rates and not specific to the case at hand.

2.4 *Type I and type II error rates*

The FRStat's algorithm estimates the type I (α) and type II (β) error rates corresponding to a 'decision' on the commonality of the source of a pair of trace and source impressions in a case at hand based on their observed level of similarity $d = \delta(t, c)$. As in the classical hypothesis testing framework, these error rates are 'tail probabilities' assigned using the sampling distributions mentioned earlier.

The rate of erroneous exclusions, α, corresponds to the probability of observing similarity scores 'lower than' d in the mated sampling distribution. Formally, this corresponds to

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$\alpha = \Pr(\delta(T,C) < d|H_0)$ and is represented in light grey in Fig. 1b. Conceptually, α corresponds to the chance of observing weak levels of similarity between pairs of impressions, even if they originate from the same finger.

The rate of erroneous identifications, β, corresponds to the probability of observing scores 'higher than' d in the non-mated sampling distribution. Formally, this corresponds to $\beta = \Pr(\delta(T,C) > d|H_1)$ and is represented in dark grey in Fig. 1b. Conceptually, β corresponds to the chance of observing 'high' levels of similarity between pairs of non-mated impressions.

In the classical hypothesis testing framework, a high value for α should lead to the 'decision' that two considered impressions are from the same source, while a high value for β should lead to the 'decision' that the two impressions are not from the same source.

2.5 *FRStat operational use and possible interpretation*

From my understanding of the FRStat operating procedures at the DFSC, FRStat is only used when an examiner believes that there is an 'association' between a pair of trace and control impressions. FRStat is not used in cases when the examiner believes that the examination is inconclusive or would result in an exclusion. The goal of FRStat is to quantify only the significance of associations (Swofford *et al*., 2018, section 3.6.3).

Figure 1b represents the result of a typical FRStat calculation. At first, the level of similarity between a trace, t, and a control impression, c, in a given case is calculated based on the 'sole corresponding features annotated by the fingerprint examiner' on the images of the trace and control impressions. This level of similarity is the 'observed score', $d = \delta(t,c)$. This score acts as the statistic and is used to test the pair of hypotheses introduced in Section 3.1 regarding the commonality of the source of t and c.

Secondly, FRStat also estimates the probability of type I and type II errors as discussed in Section 3.4. The rate of false exclusions, α, can be interpreted as a proxy for the chance that the fingerprint examiner conducting the examination would have to make an erroneous exclusion if that examiner was to 'decide' that the considered pair of impressions are from 'different' individuals. Similarly, the rate of false identifications, β, can be interpreted as a proxy for the chance that the fingerprint examiner conducting the examination would have to make an erroneous identification if that examiner was to 'decide' that the considered pair of impressions are from the 'same' individual. It is critical to note that type I and type II errors are associated with a 'decision' of the examiner in favour (or against) one of the hypotheses regarding the source of the impressions. Type I and II error rates are not meant to provide some 'measure of support' for one of the hypotheses. These error rates can only be interpreted as the chance that the examiner has made an error once a decision of identification/exclusion has been made.

Finally, FRStat returns the ratio of the two types of errors, calculated as $\frac{\alpha}{\beta}$. In Section 4.1, I explain that the authors of Swofford *et al*., (2018) mistakenly attribute to their number some properties commonly associated with likelihood ratios, and that their overall interpretation of this number is flawed. An intuition behind the FRStat number could be that, since a high value for the ratio implies that the potential for false exclusions, α, is much higher than the potential for false identifications, β, it should be 'safe' to reach the decision that the considered trace and control impressions were made by the same donor. Conversely, a lower value of the ratio could be interpreted as a sign that the observed level of correspondence between the considered trace and control impressions is such that it would lend to an unacceptable risk of erroneous identification, and therefore, that it would be

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preferable to deem the examination inconclusive. Once again, it is worth emphasizing that α, β and their ratio can only be interpreted in the context of a 'decision' of the examiner with respect to the source of a pair of trace and control impressions.

3. Issues preventing the use of FRStat in casework

The developers of FRStat make it very clear in their paper (Swofford *et al*., 2018, p. 118) that they had no intention to develop a tool capable of assigning likelihood ratios. In this article, I accept the premise that they propose a novel strategy to provide some quantitative measures of the probative value of fingerprint evidence. Nevertheless, there are many major issues associated with the FRStat algorithm that need to be exposed in order to avoid potential dramatic effects on the justice system. More specifically:

(1) There are critical discrepancies between the statistical objects that the FRStat algorithm produces, the logical and mathematical properties that they are assumed to possess and the way the DFSC proposes to report them.
(2) The statistical modelling underpinning FRStat calculation is inappropriate.
(3) The results of the validation of FRStat's performances need to be correctly interpreted in the context of the intended goals pursued by the developers of the tool.

3.1 Properties of the FRStat ratio

While the authors of Swofford *et al*. (2018, p. 118) are very clear that their ratio is not a likelihood ratio, they erroneously assume that it possesses some of the properties of the likelihood ratio:

> Equation 3: Ratio of equations 1 and 2 indicating the relative support of a given similarity statistic, GSS(t), in terms of one proposition (mated) over another (non-mated). (Swofford *et al*., 2018, caption of eq. 3, p. 116)

> From equation 3, values greater than 1 indicate a higher probability of the observed similarity statistic value among mated sources compared to non-mated sources and values less than 1 indicate a higher probability of the observed similarity statistic values among non-mated sources compared to mated sources. Values equal to 1 indicate equal probability of the observed similarity statistic value among mated and non-mated sources. (Swofford *et al*., 2018, pp. 116–118)

Ignoring the confusion between concepts such as likelihood and probability for the time being, FRStat developers seem to consider that their ratio can provide support for one hypothesis versus another, and that it has a 'tipping point' equal to 1. These are clearly properties of likelihood ratios.

I explain in Sections 3.4 and 3.5 that the numbers calculated by FRStat can only be interpreted in the context of a 'decision' of identification/exclusion made by an examiner comparing a trace and control impressions. Type I and II error rates are essential tools of classical hypothesis testing but are not meant to provide 'support' in favour (or against) a hypothesis of common source. As mentioned in Section 3.5, it seems intuitive that a large ratio (i.e. high rate of erroneous exclusions over a low rate of erroneous identifications) would tend to favour the notion that a pair of impressions was made by the same person. However, it is logically incorrect to imply that an FRStat ratio above

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one 'supports' the hypothesis of common source, while a ratio below one 'supports' the alternative hypothesis (Swofford *et al*., 2018, Equation 3). In fact, there is no rule that states that the 'tipping point' of the ratio of the rates of erroneous identifications and exclusions should be one. Such a rule would imply that an erroneous exclusion is as serious as an erroneous identification. Any attempt to set the value of the 'tipping point' of FRStat would require a general discussion between the scientific, legal and political communities in order to determine the appropriate relative severity between these two rates.[6]

It is possible to compare the FRStat ratio with a proper likelihood ratio using a toy example (Neumann *et al*., 2020).[7] Some results obtained using a toy example are reproduced in Fig. 2.

The left column of Fig. 2 compares the reported FRStat number (y-axis) to the likelihood ratio (x-axis) when the sets of observations truly originate from the same source, while the right column compares these numbers when the sets of observations truly originate from different sources. The first row of Fig. 2 compares the ratios when the characteristics of the suspected source are common and variable; in the second row, the characteristics of the suspected source are rare and variable; and in the last row, the characteristics of the suspected source are common and have virtually no variability. Figure 2 shows that in no circumstance are the FRStat number and the likelihood ratio equal. In fact, as a general rule, the FRStat number overstates the weight of the evidence, and, in most situations, the number reported by FRStat would be highly misleading in favour of the prosecution's case. This may not be of any importance when a pair of latent and control impressions truly originates from the same donor; however, this behaviour will have dramatic consequences for innocent suspects.

3.2 Compatibility with the DFSC language

The DFSC language from 2017 (U.S. Department of the Army, Defense Forensic Science Center, 2017) has several issues. The first issue is that it abuses the notion of probability. A probability can only be assigned to a numerical random variable when that variable is discrete. However, the 'amount correspondence' considered by the DFSC is a real number, at least as calculated by FRStat (Swofford *et al*., 2018),[8] and would always be associated with a probability of zero. Thus, at the very least, the DFSC language should consider the 'probability density' of the amount of correspondence. However, since the amount of correspondence between a trace and a control impression in a given case is fixed, and this fixed value is considered under two different hypotheses, the DFSC language should really be using the notion of 'likelihood'.[9]

[6] A parallel can be made between the ratio of the rates of erroneous identifications and exclusions and the risks of convicting an innocent and freeing a guilty. The determination of the relative severity in the latter case is an unsolvable problem.

[7] Once again, I understand that the developers of FRStat did not intend to assign likelihood ratios; however, since the likelihood ratio is the only coherent and logical way to quantify the 'support' provided by a piece of evidence for one of two mutually exclusive hypotheses, and that Swofford *et al*. claim that their ratio has this property, it is reasonable to compare both ratios for a same piece of evidence.

[8] Technically, it is possible to consider that the 'amount of correspondence' is an 'event', in which case it would be appropriate to talk about probability. It could also be possible to consider that the 'amount of correspondence' could be returned as a discrete variable (e.g. the number of minutiae in agreement), in which case, it would also be appropriate to talk about probability. However, as we will see further, these technicalities are moot since the probabilities mentioned in the DFSC language do not correspond to the probabilities calculated by FRStat regardless of which term is used.

[9] See Lindley (2007) for a non-mathematical introduction to these terms.

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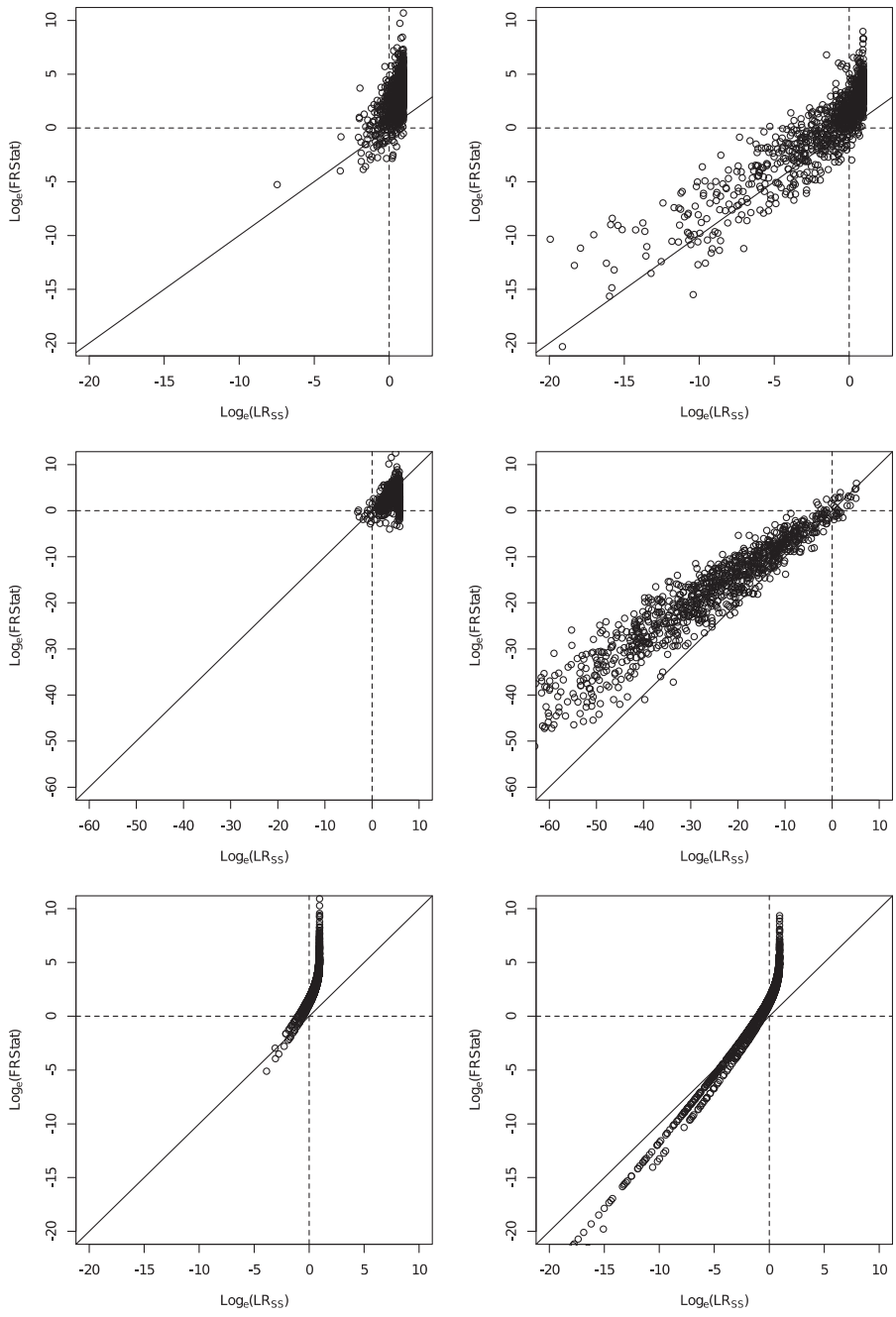


FIG. 2. Comparisons between FRStat-like values with LRs in a specific source scenario (Ommen *et al*., 2017; Ommen and Saunders, 2018). Columns: The left column reports the results when the observations are sampled when H_0 is true; the right column reports the results under H_1. First row: the source of the control impression is common and has some variance. Second row: the source of the control impression is rare and has some variance. Third row: the source of the control impression is common and has virtually no variance. The data originate from 1000 simulations using normal distributions, with which it is trivial to calculate FRStat-like numbers and compare them to the corresponding true likelihood ratios.

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Secondly, the DFSC language is expressing what has been called a 'score-based likelihood ratio'. Score-based likelihood ratios have been proposed for fingerprint evidence by numerous authors for more than a decade (Gonzalez-Rodriguez *et al*., 2005; Neumann *et al*., 2007). The likelihood ratio traditionally advocated by the legal and scientific communities is essentially a function of the similarity between the characteristics of the trace and control objects in its numerator and a function of the rarity of the trace characteristics in its denominator. In contrast, a score-based likelihood ratio is a function of the similarity between the characteristics of the trace and control objects in both its numerator and denominator as illustrated in Fig. 1c. The main shortcoming of score-based likelihood ratios is that they fail to account for the rarity of the trace characteristics,[10,11] and that they are not adequate measures of support for a hypothesis over another in the forensic context.

Finally, and most importantly, despite FRStat's results being reported using the 2017 DFSC language, the numbers produced by FRStat are not compatible with this language. I explained previously that FRStat returns three numbers for any given pair of trace and control impressions (and for the associated measure of similarity between them): the rate of erroneous exclusions (α), the rate of erroneous identifications (β) and the ratio between these two rates (α/β). The two rates 'tail probabilities' as displayed in Fig. 1b. Neither of the rates calculated by FRStat is the probability (or likelihood, or probability density) of observing 'exactly' the level of correspondence in the case at hand as the 2017 DFSC language implies it. Graphically, the DFSC language attempts to convey the situation presented in Fig. 1c, which is quite different from the FRStat calculations represented in Fig. 1b.

Formally, given observations made on a trace, t, and on a control impression, c, which have a level of similarity of $d = \delta(t, c)$:

(1) The ratio that truly has the properties assumed by Swofford *et al*. (2018) for their own ratio is $\frac{f(c,t|H_o)}{f(c,t|H_1)}$.
(2) The ratio implied by the language used by the DFSC is $\frac{f(\delta(t,c)|H_o)}{f(\delta(t,c)|H_1)}$.
(3) The ratio truly calculated by FRStat is $\frac{\Pr(\delta(T,C)<d|H_0)}{\Pr(\delta(T,C)>d|H_1)}$.

In summary, FRStat produces a mathematical object, which is interpreted using the property of another one, and reported using a language corresponding to a third one.

[10] An example based on serology data illustrates the difference. Assume the distribution of ABO blood types in the USA is 44% O; 42% A; 10% B; and 4% AB. Further assume that a suspect has the same blood type as a bloodstain recovered at the crime scene. Then (1) the probability of observing 'corresponding features' (the blood types) in two blood samples if they both come from the same person is 1 regardless of the blood type; (2) the probability of observing 'corresponding features' in two blood samples if they come from two different individuals is $0.44^2 + 0.42^2 + 0.1^2 + 0.04^2 = 0.3816$. Note that neither of the two probability statements is conditioned on the blood type actually observed. As a result, the 'weight of evidence' for the corresponding blood types is $1/0.3816 = 2.62$.

However, since the serologist observes a specific blood type, the weight of evidence should be calculated by accounting for the varying rarity of the different blood types. The probability statements of interest should really be: (1) the probability of observing a bloodstain and a suspect with blood type XXX if the bloodstain was made by the suspect; the probability of observing a bloodstain and a suspect with blood type XXX if the bloodstain was made by somebody else than the suspect. The result is then O: $1/0.44 = 2.27$; A: $1/0.42 = 2.38$; B: $1/0.10 = 10$; and AB: $1/0.04 = 25$. Therefore, the 'correspondence' ratio of 2.62 overstates the weight of the evidence if the suspect's blood and the crime scene bloodstain are common (types O and A). For those types, it makes the evidence appears stronger than it is. Contrariwise, 2.62 understates the weight of the evidence if the blood types are the rare (types B and AB). It makes the evidence appears wearker than it is.

[11] An extended discussion of limitations of score-based likelihood ratios will be the subject of a second paper and is summarized in Neumann *et al*. (2020).

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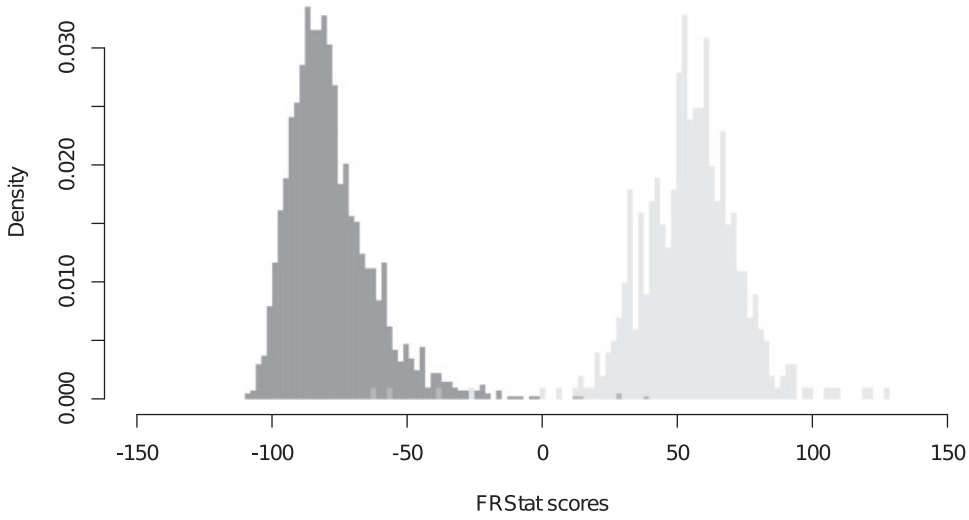


FIG. 3. Minor differences between these figures are due to the use of different bin sizes for the histograms. Same source scores ($N = 499$) are shown in light grey (right distribution). Different sources scores ($N = 2000$) are shown in darker grey (left distribution). The figure shows two trailing tails of low similarity scores between same source impressions and high similarity scores between different sources impressions. This figure is reproduced from fig. 5 of Swofford *et al.* (2018).

3.3 Statistical underpinning of the FRStat algorithm[12]

FRStat algorithm estimates the rates of false exclusions and false identifications by modelling the tail of the probability distributions of the scores in the mated and non-mated datasets (Section 3). Figure 5 in Swofford *et al.* (2018) shows that these two distributions are well-separated upwards of nine features, but that the non-mated score distribution has a consistent trailing right tail, indicating that high score values for non-mated pairs of impressions are not uncommon. The plot for 15 features in fig. 5 from Swofford *et al.* (2018) is reproduced below as Fig. 3.

FRStat is intended to be primarily used when an examiner believes that two impressions are sufficiently similar that the common source hypothesis cannot be excluded. Thus, any given comparison processed through the FRStat algorithm is likely to return a high similarity score, regardless of the true origin of the impressions. To avoid implicating an innocent suspect and, by extension, executing miscarriages of justice, it is critical that FRStat be able to provide accurate information on how often such high scores are expected to be observed when the pairs of impressions are not from the same source. For example, Fig. 3 shows that there is a non-trivial amount of non-mated high scores with 5 out of 2000 (0.25%) scores being larger than 0 for 15 features. The point is that the modelling of the mated score distribution (henceforth the rate of false exclusions) is not that critical overall, as opposed to the modelling of the non-mated score distribution, and in particular of its right tail, which is used to assign the rate of erroneous identifications.

The authors of Swofford *et al.* (2018) have chosen to model their score distributions using mixtures of logistic distributions. Logistic distributions are unimodal and symmetrical distributions that

[12] All data and information supporting the statistical analysis in this section were obtained as part of the discovery material in the Ex Parte Areli Escobar case (Ex Parte Areli Escobar, 2019). The data are available to the author of this paper but should be requested directly to the authors of Swofford *et al.* (2018).

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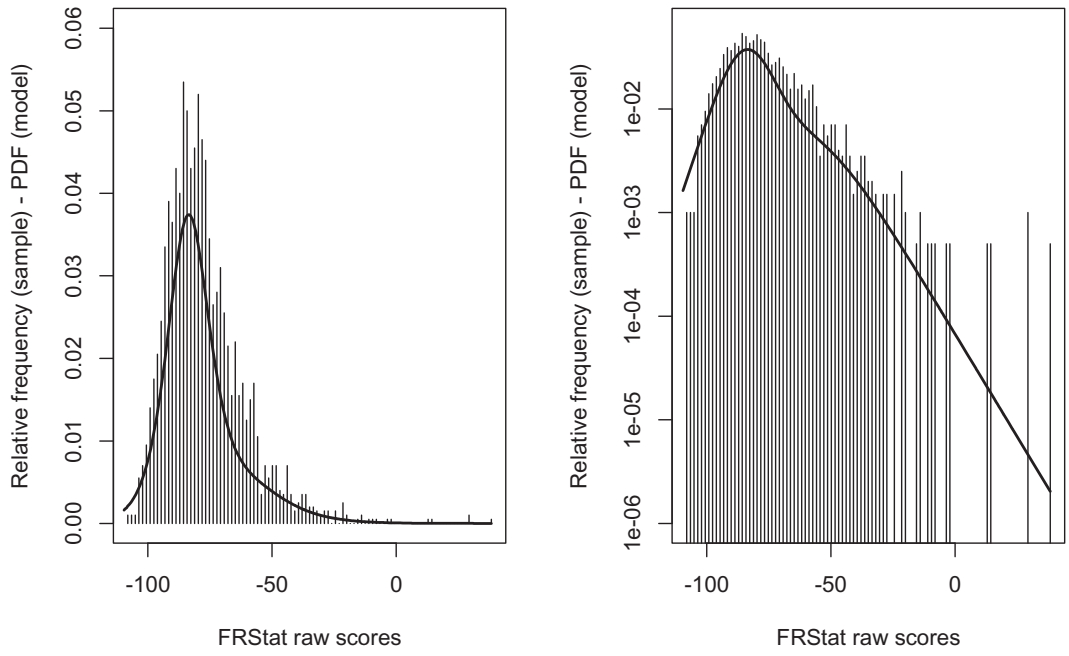


FIG. 4. Comparison between the observed distribution of non-mated scores for 15 features (histogram) and the mixture of logistic distributions chosen to represent the data (solid line). The parameters of the mixture of two logistic distributions for the non-mated scores for 15 features used by Swofford *et al.* (2018) are: weights: 0.8, 0.2; locations: −83.75, −61.25; and scale: 5.625, 10.9375. The relative frequency of the scores is shown in absolute scale (left panel) and logarithmic scale (right panel).

have fatter tails than Gaussian distributions and will give higher probability density to rarer observations (hence making them appear less rare). The general idea behind the choice of using logistic distributions seems to be that fatter tails will provide larger estimates for the rate of erroneous identifications and, therefore, will consistently result in smaller FRStat ratios. This strategy can be perceived as more favourable to an accused. The key question is whether this strategy is sufficient to represent the true rate at which larger similarity scores can be observed between pairs of impressions that are not from the same source.

Figure 4 shows the model chosen by Swofford *et al.* (2018) to represent the distribution of scores for the non-mated distribution of scores for 15 features used in FRStat and represented in Fig. 3, together with a histogram of the data used to estimate the parameters of the model (the data are the same as in Fig. 5 above). The left panel of Fig. 4 superimposes the model (solid line) and the data (histogram), and it appears that the model is reasonably appropriate for the body of the data. The right panel of Fig. 4 shows the same data as in the left panel using a logarithm transform on the *y*-axis. This logarithm transform allows to better observe data points that have a rarer occurrence. The right panel shows that while the probability density function 'dives' in the right rail, the relative frequency of the actual observations remains steady, even for fairly high similarity scores between non-mated pairs of impressions. This discrepancy between the model and the data ultimately results in tail probabilities that significantly underestimate the rate of erroneous identifications in casework.

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TABLE 1 *Comparison between expected number of scores between pairs of impressions originating from different donors (with 15 features in common) and the relative frequency of scores observed by Swofford's et al. (2018) paper*

DS scores greater than	0	25	50
FRStat expected number of scores based on chosen model	73 in 100 000 (0.073%)	7 in 100 000 (0.007%)	0.7 in 100 000 (0.0007%)
Observed number of scores	1300 in 100 000 (35 in 2694 or 1.29%)	519 in 100 000 (14 in 2694 or 0.51%)	111 in 100 000 (3 in 2694 or 0.11%)

Notes: The numbers in parentheses are the numbers of scores actually observed. They have been rescaled to a common scale for easier comparison between expected and observed scores.

In addition to Fig. 4, it is possible to compare the expected number of scores based on the mixture of logistic distributions used by FRStat and the actual number of scores observed by Swofford *et al.* (2018) during their experiments. Table 1 compares these expected and observed numbers of scores for the non-mated distribution of scores for 15 features. The observed number of scores for 15 features includes all scores calculated by Swofford *et al.* (2018) to support fig. 5 and tables 4, 5a and 5b of their paper. This represents all non-mated scores obtained for 15 features by the research team and reported in their paper.[13]

Table 1 shows that the mixture of logistic distributions significantly underestimates the number of high similarity scores between pairs of impressions from different donors. This means that FRStat fails to appropriately account for the probability of observing high similarity scores between a latent print and the control impression from an innocent suspect. This is particularly worrying given that FRStat is only meant to be used when an examiner cannot exclude a person as the source of a latent impression, which will only happen in case of highly similar prints. In these cases, the current modelling of the tails of the non-mated score distributions by FRStat will be extremely prejudicial to potentially innocent suspects.

There are three main issues that explain the lack of suitability of the FRStat mixture of logistic distributions to represent the tails of the score data. The first issue is that Swofford *et al.* collected non-mated scores by randomly associating latent and control impressions (Swofford *et al.*, 2018, p. 155, see footnote 4 above) instead of selecting pairs of highly similar non-mated impressions. The second issue is that, despite having fatter tails than a Gaussian distribution, the tails of the logistic distribution are still governed by the natural exponential function, which rapidly drives the densities in the tail to zero for values of the random variable that are away from the mean. Thus, this distribution is not adequate to model the long flat tails that are observed in the data (Figs. 3 and 4). The last issue is that Swofford *et al.* (2018) rely on an inappropriate test to assess the quality of the fit of the mixtures of logistic distributions. Swofford *et al.* (2018) learn the parameters of the logistic

[13] Figure 5 in Swofford *et al.* 2018 has 5 out of 2000 scores above 0; table 4 has 0 out of 500 scores above 0; table 5a has 6 out of 100 scores above 0 and table 5b has 24 out of 100 scores above 0. Some scores up to 54 have been observed. Combining all these scores collected under different conditions (see footnote 4 in Section 2.3) may be questionable. However, (1) this is the best estimate of the relative frequency of high similarity scores for non-mated pairs of impressions that one can have based on Swofford *et al.* (2018), and (2) the present exercise is merely intended to indicate that, in the typical casework situation where FRStat is intended to be used, FRStat tail modelling will vastly underestimate the expected occurrence of large similarity scores between non-mated pairs of impressions (in other words, this exercise is not meant to quantify the lack of fit).

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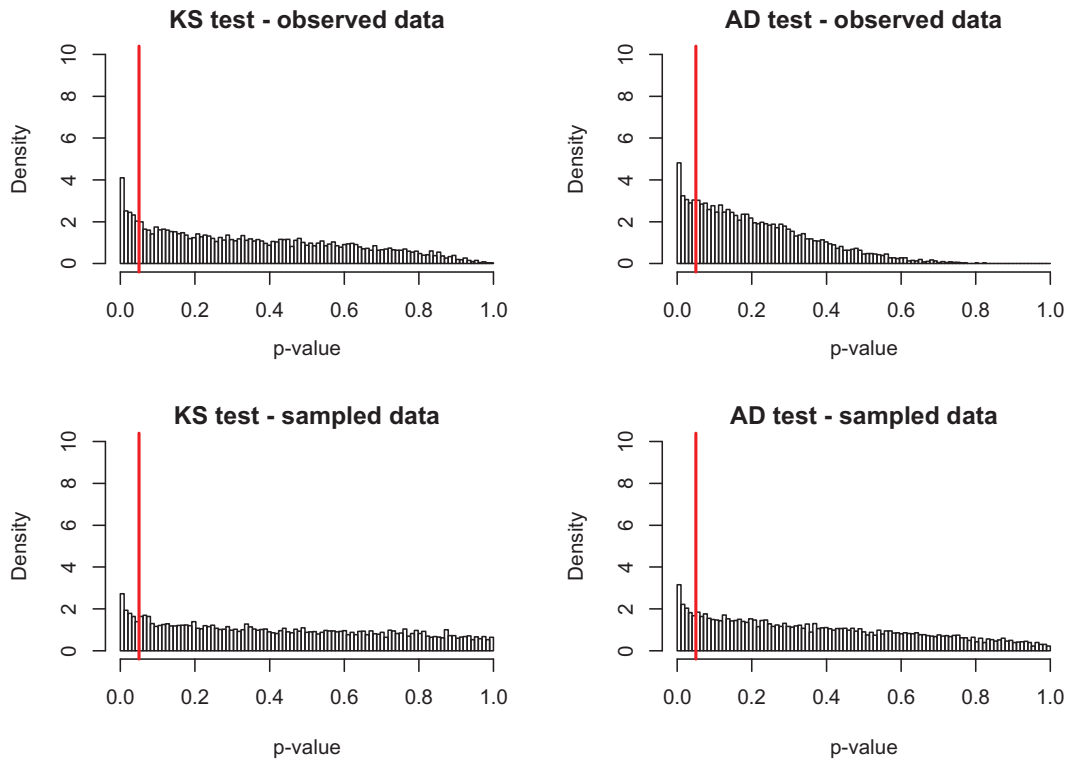


FIG. 5. Results from the simulations comparing the distributions of the p-values obtained using the KS test and the AD test. The vertical line indicates the significance level of 0.05 used by Swofford *et al.* (2018).

distributions for the mated and non-mated datasets for each number of features (from 5 to 15) by maximum likelihood estimation using 75% of the score data. The quality of the fit between the observed similarity scores and the parametric model is tested on the remaining 25% of the score data using the Kolmogorov–Smirnov (KS) distance between the empirical and parametric distributions. The equation of the KS test statistics for a proposed cumulative distribution function $F(x)$ and an empirical distribution $F_n(x)$ of n observations is:

$$D_n = \sup_x |F_n(x) - F(x)|$$

It is clear this test statistic is driven by the maximum absolute distance between the proposed and observed distributions. This maximum absolute distance will always live in the body of the distribution and never in the tails. This explains the insensitivity of the KS test to the lack of fit in the tails of the distributions. Swofford *et al.* (2018) could have used a different distance, such as the Anderson–Darling (AD) distance:

$$A_n^2 = \int_{-\infty}^{\infty} \frac{(F_n(x) - F(x))^2}{F(x)(1 - F(x))} dF(x)$$

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The AD statistic is a weighted average of all differences between the proposed and observed distributions. Mechanically, the AD statistic gives heavier weights to differences between the tails of the observed and proposed distributions.[14] The lack of power of the KS test can be seen in Fig. 5. It is a well-known fact that the distribution of the *p*-value for a statistical test, when the null hypothesis is true, is uniform. Figure 5 compares empirical distributions of the *p*-values for four different tests obtained by simulation using the score data supporting FRStat algorithm. Each simulation ran as follows:

(1) A dataset containing 75% of the non-mated score data from FRStat was created by randomly sampling from the same 2000 observed scores as Swofford *et al*. (2018).
(2) The parameters of the mixture of logistic distributions for the score distribution were learned by the maximum likelihood estimate.
(3) A KS test was performed using the proposed mixture of distributions and the remaining 25% of the observed scores. The *p*-value was stored.
(4) An AD test was performed using the proposed mixture of distributions and the remaining 25% of the observed scores. The *p*-value was stored.
(5) A dataset of 1500 scores was resampled from the proposed mixture of distributions obtained in two;
(6) A KS test was performed using the proposed mixture of distributions and the scores resampled from that mixture. The *p*-value was stored.
(7) An AD test was performed using the proposed mixture of distributions and the scores resampled from that mixture. The p-value was stored.
(8) Steps 1–7 were repeated 1000 times.

Steps 1 through 3 reproduce exactly what Swofford *et al*. (2018) did. Step 4 performs the same test as they did, but with the more powerful AD statistic. Steps 5–7 aim at studying the distributions of the *p*-values when the null hypothesis that the observed data truly originate from the proposed distribution is true with both tests. Since the dataset supporting FRStat is finite, repeating the simulation from Steps 1–7 does not result in independent *p*-values. Therefore, some departure from uniformity is expected. The upper left panel in Fig. 5 shows the distribution of the *p*-values for Step 3. This distribution represents the distribution of the *p*-values from the experiment in Swofford *et al.* (2018). This panel shows some departure of uniformity (which is expected given the lack of independence of the simulations) and shows that Swofford *et al*. (2018) had a fair chance of observing a *p*-value greater than 0.05 when performing their test of adequacy. That said, the upper left panel can be compared with the upper right panel, which shows the results from the more powerful test in Step 4 of the simulations, and the lower left panel, which presents the results from Step 6. Comparing these panels indicates that the lack of uniformity of the *p*-value distribution in the upper left panel may not be the sole product of the lack of independency of the simulations (since the distribution in the lower left panel is much more uniform) but may result from the proposed and observed distributions being incompatible. It also highlights the lack of power of the KS test to detect the departures of the proposed distribution in the tails of the observed score distribution (since the null hypothesis of equal distributions is clearly wrong based on the AD test).

[14] When $F(x) = 0.5$, $F(x)(1 - F(x)) = 0.25$ and the weight is $\frac{1}{F(x)(1-F(x))} = 4$. Any other value of $F(x)$ will result in a larger weight. For example, when $F(x) = 0.99$ (i.e. probability in the right tail is 0.01), $F(x)(1 - F(x)) = 0.0099$ and the weight is $\frac{1}{F(x)(1-F(x))} = 101.01$.

3.4 Validation of the performances of the FRStat algorithm

The previous sections report the main issues associated with the calculation, interpretation and reporting of the FRStat number. From the arguments in these sections, it is clear that the FRStat number does not have the interpretation or behaviour of a likelihood ratio, that it cannot be reported using the DFSC language and that its calculation is plagued by a lack of adequacy between the chosen parametric distributions and the observed data.

Nevertheless, it is possible to consider using the FRStat number as a purely deterministic rule. For example, it is possible to consider reporting all comparisons that produce a number above 100 as identifications[15] and all others as inconclusive (with the understanding the use of FRStat is limited to comparisons when the commonality of the origin of the pair of impressions cannot be excluded). Using this rule, it is possible to estimate the rate of erroneous identifications (i.e. the number of pairs of impressions that originate from different donors and show an FRStat number greater than 100) and the rate of missed identifications (i.e. the number of pairs of impressions that originate from the same donor and show an FRStat number lesser than 100). In this context, the calculation and actual value (or magnitude) of the FRStat number are irrelevant to the rule. The only important pieces of information that need to be reported are the calculated number in the case at hand, the value of the decision threshold that defines the rule and the rates of erroneous identifications and missed identifications for that decision threshold.

Tables 1, 4, 5a and 5b in Swofford *et al*. (2018) propose such data.[16] Table 1 in Swofford *et al*. (2018) shows estimates for the rate of missed identifications for different values of the decision rule (i.e. >1, >10 and >100). Tables 4, 5a and 5b in Swofford *et al*. (2018) report the estimates for the rate of correct exclusions for these values of the decision rule (i.e. <1, <10 and <100). The estimates in Table 4 are calculated using a dataset of 20 latent prints and 25 non-mated control impressions, resulting in 500 dependent cross-comparisons. In the experiment resulting in Table 4, the 25 non-mated control impressions were not selected for their similarity with the 20 latent prints. The estimates in tables 5a and 5b are calculated using between 94 and 100 latent prints (number of latent prints varies for different numbers of features). In the experiment resulting in tables 5a and 5b, each latent impression was searched against a large fingerprint database of 'more than 100 million different fingerprints' (Swofford *et al*., 2018) and associated with the most similar non-matching candidate returned by the matching algorithm of the database.

The estimates for the rate of missed identifications in Table 1 are interesting but not all that valuable to assess the performance of FRStat. Indeed, missed identifications would be classified as inconclusive by the decision rule and most likely not reported. The estimates for the rate of erroneous identifications in Table 4 are also interesting in the sense that they are surprisingly high given the small sample size. For example, 99.4% of the 500 cross-comparisons displaying 14 features in agreement between the 20 latent prints and 25 control impressions that originate from different donors result in an FRStat number lesser than 100. This means that 3 of these 500 comparisons result in a number above 100 and would lead to erroneous identifications using the decision rule. Bearing in mind that the 20 latent impressions and the 25 control impressions were not selected for being particularly similar to each other, this ratio of 3 in 500 shows how easy it is to find

[15] 100 is only used as an example and should not be understood as the number that should be used in casework.

[16] Tables 4, 5a and 5b in Swofford *et al*. (2018) were already briefly discussed in relation to Section 3.3 and footnote 13 above.

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TABLE 2 *Reproduction of the data supporting table 5a in Swofford et al. (2018). The results presented in table 5a are extended with estimated rates of correct exclusions for three additional thresholds: 1000, 10 000 and 100 000*

Feature quantity	Number of pairs	<1	<10	<100	<1000	<10 000	<100 000
5	99	0.566	0.788	0.980	1.000	1.000	1.000
6	99	0.687	0.747	0.980	1.000	1.000	1.000
7	96	0.688	0.719	0.896	1.000	1.000	1.000
8	99	0.747	0.788	0.812	1.000	1.000	1.000
9	99	0.818	0.818	0.828	0.960	1.000	1.000
10	97	0.814	0.835	0.845	0.897	1.000	1.000
11	96	0.802	0.823	0.823	0.854	0.958	1.000
12	98	0.857	0.867	0.888	0.898	0.939	0.990
13	99	0.899	0.929	0.939	0.939	0.960	1.000
14	100	0.980	0.99	0.990	0.990	0.990	1.000
15	100	0.920	0.920	0.940	0.940	0.940	0.990

constellations of 14 features that are similar enough (according to the FRStat algorithm) to result in very large FRStat numbers.

The series of experiments resulting in tables 5a and 5b in Swofford *et al.* (2018) is very valuable as it tests FRStat in a situation that is intended to be similar to the casework situation where an examiner fails to exclude an innocent donor as the source of a latent print. The difference between tables 5a and 5b relates to the area of friction ridge skin used for the experiment: table 5a focuses on the delta region of the fingerprint pattern, while table 5b focuses on the core region of the pattern. The data supporting table 5a in Swofford *et al.* (2018) is reproduced below in Tables 2 and 3. In these tables, the results presented in table 5a are supplemented in two ways:

(1) Tables 4, 5a and 5b in Swofford *et al.* (2018) report the rates of correct exclusions for three different values of the decision rule: 1, 10 and 100. Table 2 below reports the rates of correct exclusions for three additional values of the rule: 1000, 10 000 and 100 000.
(2) Tables 4, 5a and 5b in Swofford *et al.* (2018) only present the rates of correct exclusions for the different values of the decision rule. Table 3 below presents the corresponding rates of erroneous identifications.

When latent and control impressions from different donors are selected for their high degree of similarity, Tables 2 and 3 show that the rates of erroneous identifications are far superior to the estimated average rate of erroneous identifications of the current fingerprint examination process, which is around 1% (Ulery *et al.*, 2011; Ausdemore *et al.*, 2019). In addition, Tables 2 and 3 indicate that these rates remain high until extremely high values are used for the decision threshold. For example, the estimates for the rate of erroneous identifications for configurations of 15 features are 6% at a value of the FRStat ratio of 10 000 and 1% at a value of the FRStat ratio of 100 000. The data in Tables 2 and 3, when combined with the rate of missed identifications and FRStat number ranges in Swofford *et al.* (2018)'s table 1 and fig. 7, suggest that low false positive rates can only be achieved for values of the decision threshold that are so high that the rates of missed identifications become very close to 100%. Overall, these observations reveal that FRStat would be a much more

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TABLE 3 *Estimated rates of erroneous identifications (in percentage) corresponding to the rates of correct exclusions presented in table 5a in Swofford et al. (2018) and in Table 2*

Feature quantity	Number of pairs	>1	>10	>100	>1000	>10 000	> 100 000
5	99	43.4	21.2	2.0	0.0	0.0	0.0
6	99	31.3	25.3	2.0	0.0	0.0	0.0
7	96	31.2	28.1	10.4	0.0	0.0	0.0
8	99	25.3	21.2	18.8	0.0	0.0	0.0
9	99	18.2	18.2	17.2	4.0	0.0	0.0
10	97	18.6	16.5	15.5	10.3	0.0	0.0
11	96	19.8	17.7	17.7	14.6	4.2	0.0
12	98	14.3	13.3	11.2	10.2	6.1	1.0
13	99	10.1	7.1	6.1	6.1	4.0	0.0
14	100	2.0	1.0	1.0	1.0	1.0	0.0
15	100	8.0	8.0	6.0	6.0	6.0	1.0

inefficient and risky deterministic decision rule than the opinion-based fingerprint examination process that it is meant to complement. This would naturally question its overall usefulness.

4. Conclusion

For several decades, legal and scientific scholars have argued that conclusions from forensic examinations should be supported by statistical data and reported within a probabilistic framework. Multiple models have been proposed to quantify the probative value of forensic evidence. Unfortunately, several of these models rely on *ad hoc* strategies that are not scientifically sound. The opacity of the technical jargon that is used to present these models and their results and the complexity of the techniques involved make it very difficult for the untrained user to separate the wheat from the chaff. This series of paper is intended to help forensic scientists and lawyers recognize issues in tools proposed to interpret the results of forensic examinations. This article focuses on the tool proposed by the Latent Print Branch of the U.S. DFSC and called FRStat.

In this article, I explore the compatibility of the results outputted by FRStat with the language used by the DFCS to report the conclusions of their fingerprint examinations, as well as issues with the statistical modelling and validation of the tool itself. My findings are as follows:

(1) The DFSC language does not use appropriate probabilistic language. In fact, the 2017 DFSC language abuses several notions of probability theory and is technically meaningless. In addition, the DFSC language does not reflect the numbers calculated by FRStat and cannot be used to report them. The DFSC could be replaced by the following more appropriate language:

> *The latent print on Exhibit ## and the standards bearing the name XXXX have corresponding ridge detail. At the observed level of correspondence between the exhibit and the standards, the chance of erroneous exclusions is ## times greater/smaller than the chance of erroneous identifications.*

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(2) While Swofford *et al.* (2018) are very clear that their algorithm is not intended to calculate likelihood ratios, they claim that the FRStat ratio possesses some of the same properties. In this article, I explain that this is not the case and that the FRStat ratio is neither equal to a likelihood ratio nor has the same logical properties.

(3) Formally, given observations made on a trace, t, and on a control impression, c, which have a level of similarity of $d = \delta(t, c)$, the ratio that truly has the properties assumed by Swofford *et al.* (2018) for their own ratio is $\frac{f(c,t|H_o)}{f(c,t|H_1)}$; while the ratio implied by the language used by the DFSC is $\frac{f(\delta(t,c)|H_o)}{f(\delta(t,c)|H_1)}$; and the ratio truly calculated by FRStat is $\frac{\Pr(\delta(T,C)<d|H_0)}{\Pr(\delta(T,C)>d|H_1)}$. Therefore, FRStat produces a mathematical object, which is interpreted using the property of another one, and reported using a language corresponding to a third one.

(4) Furthermore, the probability distributions underpinning the calculations made by the FRStat algorithms are not modelled appropriately. The mixtures of logistic distributions used by Swofford *et al.* (2018) may put more probability mass on larger/smaller values of their similarity score than Gaussian distributions would, but this is not enough. In particular, I found that these mixtures of logistic distributions significantly underrepresent the likelihood of observing large similarity score values between pairs of impressions made by different donors and thus are severely underestimating the rate of erroneous identifications reported by FRStat for any given comparison.

(5) Finally, Swofford *et al.* (2018) have studied the performance of FRStat. The results of this study are incomplete and poorly interpreted. Overall, these results show that the overall performance of FRStat is worse than the subjective comparison and decision-making method that it is meant to supplement. In particular, FRStat is subject to high rates of erroneous identifications, even for comparisons bearing many features in agreement (according to the FRStat algorithm).

FRStat is an interesting and innovative tool. Swofford *et al.* (2018) propose a strategy that diverges from the commonly advocated Bayesian approach. Furthermore, FRStat is encapsulated in a software package that can readily be used by fingerprint examiners. It would be interesting to widely distribute FRStat in the community to enable discussions around standard operating procedures involving fingerprint statistical models. These discussions could focus on whether statistical models have to be used before or after an opinion has been reached by an examiner; on how to handle different results arising from the use of the model on the same comparison by two examiners or on how to report statistical results to fact-finders? In addition, FRStat could be used to study the benefits and limitations of various workflows. For example, should a statistical model be used on every comparison or only on borderline ones; or more generally, how can statistical models be used to gain efficiency?

That said, my findings show that it is dangerous to use FRStat, as it stands, in casework as it will rapidly lead to miscarriages of justice.

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